FRI Natural language processing course 2025
Project report

University of Ljubljana
Faculty of Computer and Information Science

Natural language processing course project report

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Abstract

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Keywords
Keyword1, Keyword2, Keyword3 ...

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Introduction

The purpose of this project is to develop an AI assistant for the student office of the Faculty of computer science and Informatics of the University of Ljubljana.

It serves as a chatbot that can answer students' questions about the administrative aspects of study programmes, such as study regulations, as well as provide information about the work of the faculty's student office.

Related work that will aid us in the development of the project includes:

- An IFIP conference paper from 2020 by Adamopoulou and Moussiades, providing an overview of chatbot technology that might be useful in determining the appropriate methodology and approaches [1]
- An article by Lewis et al. about RAG [2]
- AN IEEE article dfrom 2017 describing chatbots for university-related FAQs, which aligns with our problem domain [3]
- A paper presenting BARKPLUG V.2, a Large Language Model (LLM)-based chatbot system built using Retrieval Augmented Generation (RAG) pipelines [4]
- A paper that implements and compares multiple RAG pipeline configurations, combining retrieval methods like Dense, MMR, and Hybrid, and chunking strategies like Semantic and Recursive. [5]
- A study proposing the use of local LLMs like Microsoft Phi3, Meta LLaMA3, and Mistral; to power a chatbot for prospective students at Nottingham Trent University's open days. [6]
- A study that compares unsupervised finetuning and RAG across a variety of knowledge intensive tasks [7]
- A survey that synthesizes 51 studies across educational goals, learning contexts, and application scenarios, covering technical components of RAG, identifying key challenges like hallucination, outdated knowledge, and limited multimodality [8]

Methods

Use the Methods section to describe what you did an how you did it – in what way did you prepare the data, what algorithms did you use, how did you test various solutions ... Provide all the required details for a reproduction of your work.

Below are LaTeXexamples of some common elements that you will probably need when writing your report (e.g. figures, equations, lists, code examples ...).

Equations

You can write equations inline, e.g. $\cos\pi = -1$, $E = m \cdot c^2$ and α, or you can include them as separate objects. The Bayes's rule is stated mathematically as:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}, \tag{1}$$

where A and B are some events. You can also reference it – the equation 1 describes the Bayes's rule.

Lists

We can insert numbered and bullet lists:

1. First item in the list.
2. Second item in the list.
3. Third item in the list.

- First item in the list.
- Second item in the list.
- Third item in the list.

We can use the description environment to define or describe key terms and phrases.

Word What is a word?.

Concept What is a concept?

Idea What is an idea?

Random text

This text is inserted only to make this template look more like a proper report. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam blandit dictum facilisis. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Interdum et malesuada fames ac ante ipsum primis in faucibus. Etiam convallis tellus velit, quis ornare ipsum aliquam id. Maecenas tempus mauris sit amet libero elementum eleifend. Nulla nunc orci, consectetur non consequat ac, consequat non nisl. Aenean vitae dui nec ex fringilla malesuada. Proin elit libero, faucibus eget neque quis, condimentum laoreet urna. Etiam at nunc quis felis pulvinar dignissim. Phasellus turpis turpis, vestibulum eget imperdiet in, molestie eget neque. Curabitur quis ante sed nunc varius dictum non quis nisl. Donec nec lobortis velit. Ut cursus, libero efficitur dictum imperdiet, odio mi fermentum dui, id vulputate metus velit sit amet risus. Nulla vel volutpat elit. Mauris ex erat, pulvinar ac accumsan sit amet, ultrices sit amet turpis.

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Figures

You can insert figures that span over the whole page, or over just a single column. The first one, Figure 1, is an example of a figure that spans only across one of the two columns in the report.

On the other hand, Figure 2 is an example of a figure that spans across the whole page (across both columns) of the report.

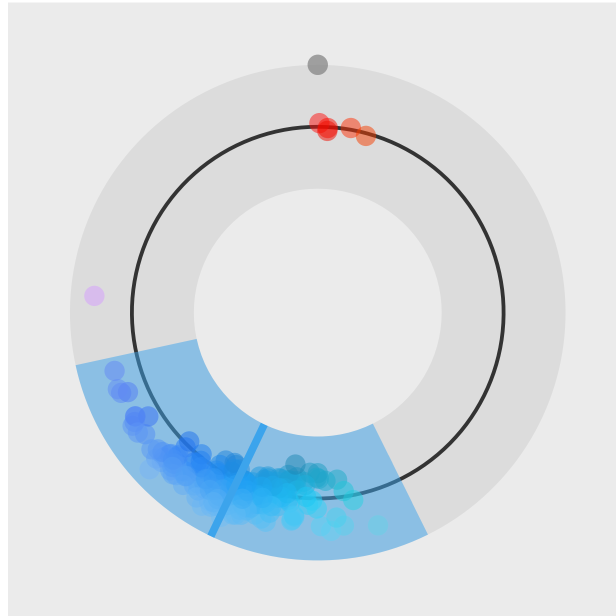

Figure 1. A random visualization. This is an example of a figure that spans only across one of the two columns.

Tables

Use the table environment to insert tables.

Table 1. Table of grades.

Name		
First name	Last Name	Grade
John	Doe	7.5
Jane	Doe	10
Mike	Smith	8

Code examples

You can also insert short code examples. You can specify them manually, or insert a whole file with code. Please avoid inserting long code snippets, advisors will have access to your repositories and can take a look at your code there. If necessary, you can use this technique to insert code (or pseudo code) of short algorithms that are crucial for the understanding of the manuscript.

Listing 1. Insert code directly from a file.

```
import os
import time
import random

fruits = ["apple", "banana", "cherry"]
for x in fruits:
  print(x)
```

Listing 2. Write the code you want to insert.

```
import(dplyr)
```

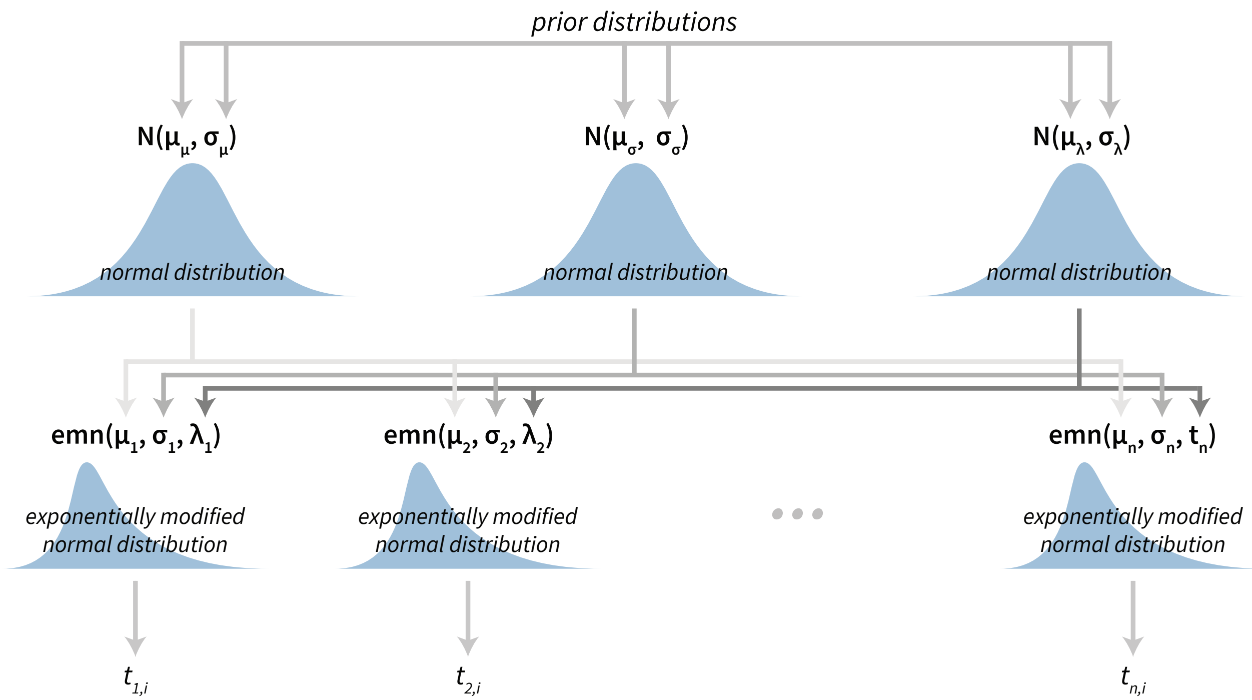

Figure 2. Visualization of a Bayesian hierarchical model. This is an example of a figure that spans the whole width of the report.

```
import(ggplot)

ggplot(diamonds,
           aes(x=carat, y=price, color=cut)) +
  geom_point() +
  geom_smooth()
```

Results

Our approach correctly obtains, processes and builds an index from the data. A demonstrative evaluation with the *Qwen/Qwen2.5-1.5B-Instruct* model on an average laptop CPU shows good results, with the model able to answer questions both in Slovene and English (see table 2), with both answers containing the expected keyword, showing that the data processing pipeline is working.

Discussion

The discrepancy in the answer generation time is most likely either a result of fluctuating system load during the evaluation or a property of the model used for answering.

Acknowledgments

Here you can thank other persons (advisors, colleagues ...) that contributed to the successful completion of your project.

References

[1] Eleni Adamopoulou and Lefteris Moussiades. An overview of chatbot technology. In *IFIP international conference on artificial intelligence applications and innovations*, pages 373–383. Springer, 2020.

[2] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems*, 33:9459–9474, 2020.

[3] Bhavika R Ranoliya, Nidhi Raghuwanshi, and Sanjay Singh. Chatbot for university related faqs. In *2017 international conference on advances in computing, communications and informatics (ICACCI)*, pages 1525–1530. IEEE, 2017.

[4] Subash Neupane, Elias Hossain, Jason Keith, Himanshu Tripathi, Farbod Ghiasi, Noorbakhsh Amiri Golilarz, Amin Amirlatifi, Sudip Mittal, and Shahram Rahimi. From questions to insightful answers: Building an informed chatbot for university resources, 2024.

[5] Maksuda Khasanova Zafar kizi and Youngjung Suh. Design and performance evaluation of llm-based rag pipelines for chatbot services in international student admissions. *Electronics*, 14(15), 2025.

Question	Expected keywords	Answer contains all expected kw.	Generation time [s]
Kako poteka prijava na izpit?	STUDIS	yes	2153.03
How do I apply for an exam?	STUDIS	yes	384.21

Table 2. Evaluation results.

[6] Akash Shanthakumar, Farhad Fassihi, Ahmad Lotfi, and Jordan Bird. *Retrieval Augmented Large Language Model Chatbots in Higher Education: A Study on University Open Days*, pages 32–44. 01 2025.

[7] Oded Ovadia, Menachem Brief, Moshik Mishaeli, and Oren Elisha. Fine-tuning or retrieval? comparing knowledge injection in llms, 2024.

[8] Zongxi Li, Zijian Wang, Weiming Wang, Kevin Hung, Haoran Xie, and Fu Lee Wang. Retrieval-augmented generation for educational application: A systematic survey. *Computers and Education: Artificial Intelligence*, 8:100417, 2025.